

Table 3. CheckV recovery on real metagenomic datasets

Dataset	Method	Complete	High-quality	Medium-quality	HQ+	MQ+	Predicted bins
Lake water (metaSPAdes)	CoCoNet	30	375	360	405	765	3,568
	vRhyme	51	463	586	514	1,100	2,540
	VAMB	107	488	666	595	1,261	10,674
	CoCoBin	0	153	33	153	186	200
	CLMB	107	423	627	530	1,157	11,993
	MetaBAT 2	119	477	856	596	1,452	9,850
	CONCOCT	3	99	12	102	114	138
	MAGE-Bin	114	783	709	897	1,606	6,649
Lake water (MEGAHIT)	CoCoNet	11	675	467	686	1,153	4,168
	vRhyme	12	392	563	404	967	2,718
	VAMB	108	390	531	498	1,029	12,533
	CoCoBin	0	176	29	176	205	222
	CLMB	104	339	490	443	933	13,856
	MetaBAT 2	103	380	649	483	1,132	11,709
	CONCOCT	2	96	12	98	110	128
	MAGE-Bin	112	656	685	768	1,453	7,811
Wastewater (metaFlye)	CoCoNet	3	349	186	352	538	996
	vRhyme	0	134	43	134	177	368
	VAMB	306	370	492	676	1,168	3,951
	CoCoBin	0	76	9	76	85	87
	CLMB	376	396	555	772	1,327	4,287
	MetaBAT 2	380	489	509	869	1,378	4,021
	CONCOCT	0	14	2	14	16	24
	MAGE-Bin	477	442	277	919	1,196	2,705
IBD (metaFlye)	CoCoNet	0	200	107	200	307	614
	vRhyme	2	100	44	102	146	254
	VAMB	237	562	341	799	1,140	3,751
	CoCoBin	0	102	12	102	114	121
	CLMB	133	432	232	565	797	2,568
	MetaBAT 2	129	289	188	418	606	2,123
	CONCOCT	3	58	1	61	62	65
	MAGE-Bin	559	610	301	1,169	1,470	3,243
Soil (metaFlye)	CoCoNet	12	2,141	1,535	2,153	3,688	9,567
	vRhyme	4	350	439	354	793	2,480
	VAMB	3,027	2,308	2,651	5,335	7,986	41,071
	CoCoBin	-	-	-	-	-	-
	CLMB	3,290	2,473	2,756	5,763	8,519	42,931
	MetaBAT 2	1,600	2,412	2,145	4,012	6,157	17,181
	CONCOCT	0	147	13	147	160	200
	MAGE-Bin	3,569	2,520	2,538	6,089	8,627	22,482

Bold indicates the highest quality-bin count within each dataset. HQ+ = Complete + High-quality; MQ+ also includes Medium-quality. Soil CoCoBin was unavailable.